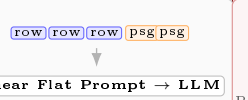
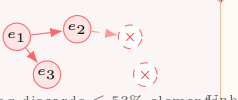
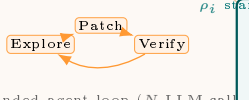
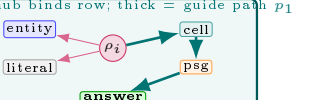
	(A) Flattened In-Context Prompting Direct LLM, Flat-RAG	(B) Online Graph + Pruning e.g., ODYSSEY (NAACL 2025)	(C) Agentic Graph Search e.g., HydraRAG (EMNLP 2025)	(D) OUEG-RAG (Ours) Training-free, Guided, Bounded
Evidence Representation				
Limitation 1 Tabular Bindings	✗ Rows flattened to text; symmetric rows get conflated	✗ Split into pairwise triples; column context is lost	✗ Table cells treated as flat entity mentions	✓ Star record hubs ρ_i guarantee row non-conflation (**Prop. 1**)
Limitation 2 Reasoning Guide	✗ None: free-form attention over the prompt	✗ Unorganized bag of triples fed to the LLM	✗ Paths found by open-ended agent turns; prone to search drift	✓ Deterministic score $\mathcal{S}(p)$ yields guide path p_1 (never filters)
Limitation 3 Loss vs. Calls	– Nothing pruned, but evidence stays unstructured	✗ Prunes ≤ 53% elements; peripheral facts are lost (39% empty)	✗ Dozens of LLM calls: high latency and API cost	✓ 100% evidence retained; bounded by two tiers (1–2 calls)
LLM Calls	**1**	**1**	**dozens**	**1 (Tier 1) or 2 (Tier 2)**

✓ Limitation addressed ✗ Limitation present – Not applicable